

In the following exercises, all coordinates and components are given with respect to a right-oriented orthonormal frame $\mathcal{K} = (O, \mathcal{B})$ where $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$.

- 4.1. Consider the vectors $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$ and $\mathbf{b} = 7\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$. Calculate $\mathbf{a} \times \mathbf{b}$.
- 4.2. Consider the points $A(1, 1, 0)$, $B(1, 2, 0)$ and $C(0, y, 0)$ for some $y \in \mathbb{R}$. Show that the area of the triangle ABC is independent of the value y . How would you argue for the similar configuration in the plane $\mathbb{E}^2$?
- 4.3. Consider the vectors $\mathbf{a}(2, -3, 1)$, $\mathbf{b}(-3, 1, 2)$ and $\mathbf{c}(1, 2, 3)$. Calculate $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$ and $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$.
- 4.4. Are the points $A(1, 2, -1)$, $B(0, 1, 5)$, $C(-1, 2, 1)$ and $D(2, 1, 3)$ coplanar?
- 4.5. Consider the tetrahedron with vertices $A(2, -1, 1)$, $B(5, 5, 4)$, $C(3, 2, -1)$ and $D(4, 1, 3)$. Determine
 - a) the volume of the tetrahedron.
 - b) the common perpendicular of the sides AB and CD .
- 4.6. Consider the vectors $\mathbf{a}(2, 3, -1)$ and $\mathbf{b}(1, -1, 3)$. Determine the vector $\mathbf{p}$ which is orthogonal to $\mathbf{a}$ and $\mathbf{b}$ and for which $\langle \mathbf{p}, 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k} \rangle = 51$.
- 4.7. Consider the points $A(1, 1, 0)$, $B(1, 0, 1)$, $C(1, -1, -1)$ and $D(0, y, z)$ for some $y, z \in \mathbb{R}$. Show that the volume of the tetrahedron $ABCD$ is independent of the values y and z .
- 4.8. Let π be a plane with normal vector $(1, 1, 1)$ intersecting the coordinate axes in A , B and C . Give formulas for the volume of the tetrahedron $OABC$ in terms of $h = d(O, \pi)$, $a = |OA|$ and $s = |AB|$ respectively.
- 4.9. Determine the matrices of the linear map $\phi : \mathbb{V}^3 \rightarrow \mathbb{V}^3$ defined by $\phi(\mathbf{v}) = (2\mathbf{i} - \mathbf{j}) \times \mathbf{v}$.
- 4.10. Consider the vectors $\mathbf{u} = -\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ and $\mathbf{w} = \mathbf{k}$. Determine the matrix of the linear map $\phi : \mathbb{V}^3 \rightarrow \mathbb{R}$ defined by $\phi(\mathbf{v}) = [\mathbf{v}, \mathbf{u}, \mathbf{w}]$.

4.1. Consider the vectors $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$ and $\mathbf{b} = 7\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$. Calculate $\mathbf{a} \times \mathbf{b}$.

$$\mathbf{a} \times \mathbf{b} = (\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}) \times (7\mathbf{i} + 4\mathbf{j} + 6\mathbf{k})$$

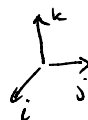
$$= \underbrace{7\mathbf{i} \times \mathbf{i}}_{=0} + 4\mathbf{i} \times \mathbf{j} + 6\mathbf{i} \times \mathbf{k} +$$

$$+ 14\mathbf{j} \times \mathbf{i} + \underbrace{8\mathbf{j} \times \mathbf{j}}_{=0} + 12\mathbf{j} \times \mathbf{k} +$$

$$- 14\mathbf{k} \times \mathbf{i} - 8\mathbf{k} \times \mathbf{j} - \underbrace{12\mathbf{k} \times \mathbf{k}}_{=0}$$

$$= -10 \underbrace{\mathbf{i} \times \mathbf{j}}_{=\mathbf{k}} + 20 \underbrace{\mathbf{i} \times \mathbf{k}}_{=-\mathbf{j}} + 20 \underbrace{\mathbf{j} \times \mathbf{k}}_{=\mathbf{i}}$$

$$= 20\mathbf{i} - 20\mathbf{j} - 10\mathbf{k}$$



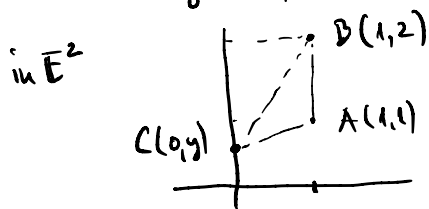
$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -2 \\ 7 & 4 & 6 \end{vmatrix} = 20\mathbf{i} - 20\mathbf{j} - 10\mathbf{k}$$

4.2. Consider the points $A(1,1,0)$, $B(1,2,0)$ and $C(0,y,0)$ for some $y \in \mathbb{R}$. Show that the area of the triangle ABC is independent of the value y . How would you argue for the similar configuration in the plane $\mathbb{E}^2$?

$$\bullet \text{ area of } \triangle ABC = \frac{1}{2} |\vec{AB} \times \vec{AC}|$$

$$\bullet \vec{AB} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \vec{AC} = \begin{bmatrix} -1 \\ y-1 \\ 0 \end{bmatrix} \quad \vec{AB} \times \vec{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 0 \\ -1 & y-1 & 0 \end{vmatrix} = \mathbf{k} \cdot \begin{vmatrix} 0 & 1 \\ -1 & y-1 \end{vmatrix} = \mathbf{k}$$

$\bullet$ the length of $\vec{AB} \times \vec{AC}$ is independent of $y \Rightarrow$ so is $\text{Area}(\triangle ABC)$



$\text{Area}(\triangle ABC) = \frac{1}{2} |\mathbf{AB}| \cdot h_c$
 where $h_c = d(C, \mathbf{AB})$ which does not depend on y since C moves parallel to $\mathbf{AB}$

4.3. Consider the vectors $\mathbf{a}(2, -3, 1)$, $\mathbf{b}(-3, 1, 2)$ and $\mathbf{c}(1, 2, 3)$. Calculate $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$ and $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$.

$$\begin{aligned}
 (\mathbf{a} \times \mathbf{b}) \times \mathbf{c} &= \langle \mathbf{a} \cdot \mathbf{c} \rangle \mathbf{b} - \langle \mathbf{b} \cdot \mathbf{c} \rangle \mathbf{a} = (2 - 6 + 3) \mathbf{b} - (-3 + 2 + 6) \mathbf{a} \\
 &= - \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} - 5 \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} -7 \\ 14 \\ -7 \end{bmatrix} \\
 \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= \langle \mathbf{a} \cdot \mathbf{c} \rangle \mathbf{b} - \langle \mathbf{a} \cdot \mathbf{b} \rangle \mathbf{c} = - \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} + 7 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 13 \\ 19 \end{bmatrix}
 \end{aligned}$$

4.4. Are the points $A(1, 2, -1)$, $B(0, 1, 5)$, $C(-1, 2, 1)$ and $D(2, 1, 3)$ coplanar?

$$\begin{aligned}
 A, B, C, D \text{ coplanar} &\Leftrightarrow \underbrace{\overrightarrow{AB}}_1, \underbrace{\overrightarrow{AC}}_2, \underbrace{\overrightarrow{AD}}_3 \text{ linearly dependent} \\
 &\Downarrow \\
 &\begin{vmatrix} -1 & -2 & 1 \\ -1 & 0 & -1 \\ 6 & 2 & 4 \end{vmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \text{Volume of tetrahedron } ABCD &= 0 \Leftrightarrow \underbrace{\begin{vmatrix} -1 & -1 & 6 \\ -2 & 0 & 2 \\ 1 & -1 & 4 \end{vmatrix}}_{12 - 2 - 8 - 2 = 0} = 0 \quad \text{true}
 \end{aligned}$$

4.5. Consider the tetrahedron with vertices $A(2, -1, 1)$, $B(5, 5, 4)$, $C(3, 2, -1)$ and $D(4, 1, 3)$. Determine

- the volume of the tetrahedron.
- the common perpendicular of the sides AB and CD .

$$a) \quad V_d = \frac{1}{6} |\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}| = \frac{1}{6} \begin{vmatrix} 3 & 6 & 3 \\ 1 & 3 & -2 \\ 2 & 2 & 2 \end{vmatrix} = 1 - 3 = 3$$

$$b) \quad \underbrace{AB}_A: \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}}_{\overrightarrow{AB}/3} + t \underbrace{\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}}_{\overrightarrow{AB}/3} \quad \underbrace{CD}_C: \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}}_C + \lambda \underbrace{\begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}}_{\overrightarrow{CD}}$$

$$\frac{\vec{AB}}{3} \times \vec{CD} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 1 & -1 & 4 \end{vmatrix} = \hat{i} \begin{vmatrix} 2 & 1 \\ -1 & 4 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & 1 \\ 1 & 4 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & 2 \\ 1 & -1 \end{vmatrix} = 3 \begin{bmatrix} 3 \\ -1 \\ -1 \end{bmatrix}$$

$$\text{common perp : } \begin{cases} \begin{vmatrix} x-2 & y+1 & z-1 \\ 1 & 2 & 1 \\ 3 & -1 & -1 \end{vmatrix} = 0 \Leftrightarrow -1(x-2) + 4(y+1) - 7(z-1) = 0 \\ \begin{vmatrix} x-3 & y-2 & z+1 \\ 1 & -1 & 4 \\ 3 & -1 & -1 \end{vmatrix} = 0 \Leftrightarrow 5(x-3) + 13(y-2) + 2(z+1) = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} -x + 4y - 7z + 13 = 0 \\ 5x + 13y + 2z - 39 = 0 \end{cases}$$

4.6. Consider the vectors $\mathbf{a}(2, 3, -1)$ and $\mathbf{b}(1, -1, 3)$. Determine the vector $\mathbf{p}$ which is orthogonal to $\mathbf{a}$ and $\mathbf{b}$ and for which $\langle \mathbf{p}, 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k} \rangle = 51$.

Sol 1

take an arbitrary vector $\mathbf{v}(x, y, z)$ and impose the

conditions: $\mathbf{v} \in \langle \mathbf{a}, \mathbf{b} \rangle^\perp \Leftrightarrow \mathbf{v} \perp \mathbf{a} \text{ and } \mathbf{v} \perp \mathbf{b}$

$$\Leftrightarrow \mathbf{v} \cdot \mathbf{a} = 0 \text{ and } \mathbf{v} \cdot \mathbf{b} = 0$$

$$\Leftrightarrow \begin{cases} 2x + 3y - z = 0 \\ x - y + 3z = 0 \end{cases}$$

this system allows us to express y and z in terms of x

then $\mathbf{v} \in \langle \mathbf{a}, \mathbf{b} \rangle^\perp \Leftrightarrow \mathbf{v} = \mathbf{v}(x, y(x), z(x))$

Sol 2

$$\langle \mathbf{a}, \mathbf{b} \rangle^\perp = \langle \mathbf{a} \times \mathbf{b} \rangle = \left\langle \begin{bmatrix} 8 \\ -7 \\ -5 \end{bmatrix} \right\rangle = \{ (8t, -7t, -5t) : t \in \mathbb{R} \}$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -1 \\ 1 & -1 & 3 \end{vmatrix} = 8\hat{i} - 7\hat{j} - 5\hat{k}$$

$$p \in \langle a, b \rangle^+ \Leftrightarrow p = t \begin{pmatrix} 8 \\ -7 \\ -5 \end{pmatrix} \text{ for some } t \in \mathbb{R}$$

$$p \cdot (2i - 3j + 4k) = 51 \Leftrightarrow (t8i - t7j - t5k) \cdot (2i - 3j + 4k) = 51$$

$$\Leftrightarrow 16t + 21t - 20t = 51$$

$$\Leftrightarrow t = \frac{51}{17} = 3$$

4.7. Consider the points $A(1, 1, 0)$, $B(1, 0, 1)$, $C(1, -1, -1)$ and $D(0, y, z)$ for some $y, z \in \mathbb{R}$. Show that the volume of the tetrahedron $ABCD$ is independent of the values y and z .

$$\cdot \text{Vol}_{\text{tetra}}(ABCD) = \frac{1}{6} |[\vec{AB}, \vec{AC}, \vec{AD}]|$$

$$\cdot \vec{AB} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \quad \vec{AC} = \begin{bmatrix} 0 \\ -2 \\ -1 \end{bmatrix} \quad \vec{AD} = \begin{bmatrix} -1 \\ y-1 \\ z \end{bmatrix}$$

$$\cdot [\vec{AB}, \vec{AC}, \vec{AD}] = \begin{vmatrix} 0 & -1 & 1 \\ 0 & -2 & -1 \\ -1 & y-1 & z \end{vmatrix} = -1 \begin{vmatrix} -1 & 1 \\ -2 & -1 \end{vmatrix} = -3$$

$$\Rightarrow \text{Vol}_{\text{tetra}}(ABCD) = \frac{1}{2} \text{ constant / independent of } y \text{ and } z$$

Obs $\text{Vol}_{\text{tetra}}(ABCD) = \underbrace{\frac{1}{3} \text{Area}(ABC)}_{\text{indep. of } y, z} \cdot \underbrace{h_D}_{d(D, ABC)} \quad \text{(as volume of pyramid with base } ABC)$

but D moves in a plane parallel to ABC
 $\Rightarrow$ this distance is constant

4.8. Let π be a plane with normal vector $(1, 1, 1)$ intersecting the coordinate axes in A, B and C . Give formulas for the volume of the tetrahedron $OABC$ in terms of $h = d(O, \pi)$, $a = |OA|$ and $s = |AB|$ respectively.

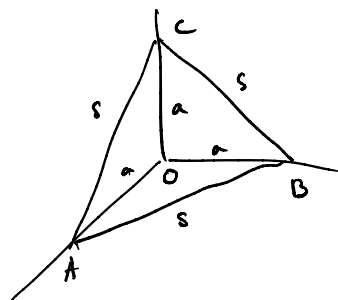
we may assume $A(a, 0)$, i.e. $a > 0$

$$\bar{1}: \frac{x}{a} + \frac{y}{a} + \frac{z}{a} = 1 \Leftrightarrow x + y + z - a = 0$$

$$h = d(O, \pi) = \frac{|a|}{\sqrt{3}} = \frac{a}{\sqrt{3}} \Rightarrow a = \sqrt{3}h$$

$$s = \sqrt{2}a \Rightarrow a = s/\sqrt{2}$$

$$V = \text{Vol}(OABC) = \frac{1}{6} \text{ volume cube with side } a$$



$$\Rightarrow V = \frac{a^3}{6} = \frac{\sqrt{3}}{2} h^3 = \frac{s^3}{12\sqrt{2}}$$

$$\text{check with Heron: Area}(ABC) = \frac{\sqrt{3}}{4} s^2 \Rightarrow V = \frac{1}{3} \cdot \frac{\sqrt{3}}{4} s^2 \cdot h = \frac{s^3}{12\sqrt{2}}$$

4.9. Determine the matrices of the linear map $\phi: \mathbb{V}^3 \rightarrow \mathbb{V}^3$ defined by $\phi(v) = (2i - j) \times v$.

$$\phi(i) = (2i - j) \times i = 2i \times i - j \times i = i \times j = k$$

$$\phi(j) = (2i - j) \times j = 2i \times j = 2k$$

$$\phi(k) = (2i - j) \times k = 2i \times k - j \times k = -2j - i$$

$$\Rightarrow [\phi] = \begin{bmatrix} \uparrow \phi(i) & \uparrow \phi(j) & \uparrow \phi(k) \\ \downarrow & \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & -2 \\ 1 & 2 & 0 \end{bmatrix}$$

4.10. Consider the vectors $u = -i + 3j + k$ and $w = k$. Determine the matrix of the linear map $\phi: \mathbb{V}^3 \rightarrow \mathbb{R}$ defined by $\phi(v) = [v, u, w]$.

$$\phi(i) = \begin{vmatrix} 1 & 0 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 3 \quad \phi(j) = \begin{vmatrix} 0 & 1 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 1 \quad \phi(k) = \begin{vmatrix} 0 & 0 & 1 \\ -1 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 0$$

$$\Rightarrow [\phi] = \begin{bmatrix} \uparrow \phi(i) & \uparrow \phi(j) & \uparrow \phi(k) \\ \downarrow & \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} 3 & 1 & 0 \end{bmatrix}$$